



H2 Standard integrals

Name: _____

Class: _____

Date: _____

Time: **507 minutes**

Marks: **420 marks**

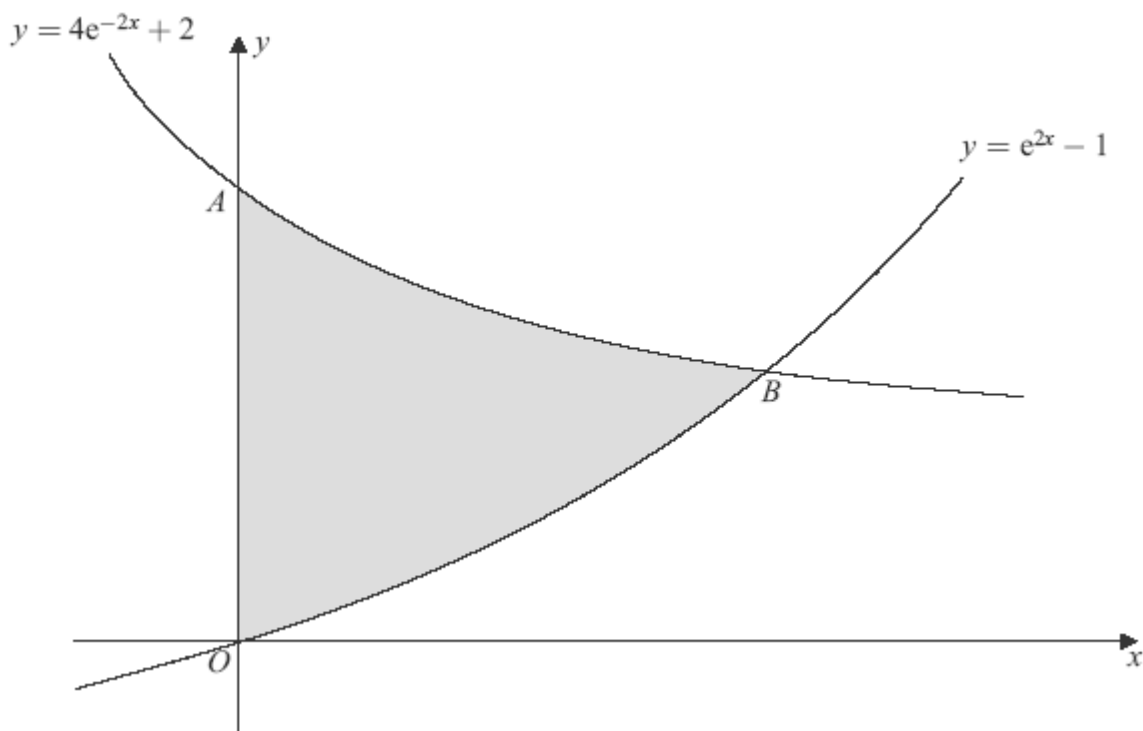
Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

The diagram shows the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$.



The curve $y = 4e^{-2x} + 2$ crosses the y -axis at the point A and the curves intersect at the point B .

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} - 1$.

(4)

- (b) Write down the coordinates of the point A .

(1)

- (c) (i) Show that the x -coordinate of the point B satisfies the equation

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

(2)

- (ii) Hence find the exact value of the x -coordinate of the point B .

(3)

- (d) Find the exact value of the area of the shaded region bounded by the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$ and the y -axis.

(5)

(Total 15 marks)

Q2.

- (a) (i) Express $\frac{7x-3}{(x+1)(3x-2)}$ in the form $\frac{A}{x+1} + \frac{B}{3x-2}$. (3)

- (ii) Hence find $\int \frac{7x-3}{(x+1)(3x-2)} dx$. (2)

- (b) Express $\frac{6x^2 + x + 2}{2x^2 - x + 1}$ in the form $P + \frac{Qx + R}{2x^2 - x + 1}$. (3)
- (Total 8 marks)

Q3.

- (a) Solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x+1)^{\frac{1}{2}}$$

given that $x = 80$ when $t = 0$. Give your answer in the form $x = f(t)$. (6)

- (b) A fungus is spreading on the surface of a wall. The proportion of the wall that is unaffected after time t hours is $x\%$. The rate of change of x is modelled by the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x+1)^{\frac{1}{2}}$$

At $t = 0$, the proportion of the wall that is unaffected is 80%. Find the proportion of the wall that will still be unaffected after 60 hours. (2)

- (c) A biologist proposes an alternative model for the rate at which the fungus is spreading on the wall. The total surface area of the wall is 9 m^2 . The surface area that is **affected** at time t hours is $A \text{ m}^2$. The biologist proposes that the rate of change of A is proportional to the product of the surface area that is affected and the surface area that is unaffected.

- (i) Write down a differential equation for this model.

(You are not required to solve your differential equation.) (2)

- (ii) A solution of the differential equation for this model is given by

$$A = \frac{9}{1 + 4e^{-0.09t}}$$

Find the time taken for 50% of the area of the wall to be affected. Give your answer in hours to three significant figures.

(4)

(Total 14 marks)

Q4.

(a) The polynomial $p(x)$ is given by $p(x) = x^3 + x - 10$.

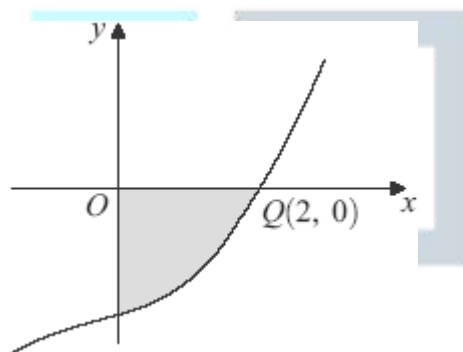
(i) Use the Factor Theorem to show that $x - 2$ is a factor of $p(x)$.

(2)

(ii) Express $p(x)$ in the form $(x - 2)(x^2 + ax + b)$, where a and b are constants.

(2)

(b) The curve C with equation $y = x^3 + x - 10$, sketched below, crosses the x -axis at the point $Q(2, 0)$.



(i) Find the gradient of the curve C at the point Q .

(4)

(ii) Hence find an equation of the tangent to the curve C at the point Q .

(2)

(iii) Find $\int (x^3 + x - 10) dx$.

(3)

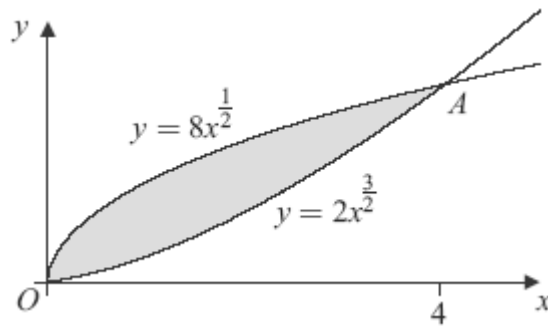
(iv) Hence find the area of the shaded region bounded by the curve C and the coordinate axes.

(2)

(Total 15 marks)

Q5.

The diagram shows a sketch of the curves with equations $y = 2x^{\frac{3}{2}}$ and $y = 8x^{\frac{1}{2}}$.



The curves intersect at the origin and at the point A , where $x = 4$.

- (a) (i) For the curve $y = 2x^{\frac{3}{2}}$, find the value of $\frac{dy}{dx}$ when $x = 4$. (2)
- (ii) Find an equation of the normal to the curve $y = 2x^{\frac{3}{2}}$ at the point A . (4)
- (b) (i) Find $\int 8x^{\frac{1}{2}} dx$. (2)
- (ii) Find the area of the shaded region bounded by the two curves. (4)
- (c) Describe a single geometrical transformation that maps the graph of $y = 2x^{\frac{3}{2}}$ onto the graph of $y = 2(x+3)^{\frac{3}{2}}$. (2)

(Total 14 marks)

Q6.

- (a) A differential equation is given by $\frac{dx}{dt} = -kt e^{\frac{1}{2}x}$, where k is a positive constant.
- (i) Solve the differential equation. (3)
- (ii) Hence, given that $x = 6$ when $t = 0$, show that $x = -2 \ln\left(\frac{kt^2}{4} + e^{-3}\right)$.

(3)

- (b) The population of a colony of insects is decreasing according to the model

$$\frac{dx}{dt} = -kt e^{\frac{1}{2}x}, \text{ where } x \text{ thousands is the number of insects in the colony after time } t \text{ minutes. Initially, there were 6000 insects in the colony.}$$

Given that $k = 0.004$, find:

- (i) the population of the colony after 10 minutes, giving your answer to the nearest hundred;
- (ii) the time after which there will be no insects left in the colony, giving your answer to the nearest 0.1 of a minute.

(2)

(2)

(Total 10 marks)

Q7.

- (a) Write down the value of n given that $\frac{1}{x^4} = x^n$.

(1)

- (b) Expand $\left(1 + \frac{3}{x^2}\right)^2$.

(2)

- (c) Hence find $\int \left(1 + \frac{3}{x^2}\right)^2 dx$.

(3)

- (d) Hence find the exact value of $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx$.

(2)

(Total 8 marks)

Q8.

- (a) Solve the differential equation

$$\frac{dx}{dt} = \frac{150 \cos 2t}{x}$$

given that $x = 20$ when $t = \frac{\pi}{4}$, giving your solution in the form $x^2 = f(t)$.

(6)

- (b) The oscillations of a 'baby bouncy cradle' are modelled by the differential equation

$$\frac{dx}{dt} = \frac{150 \cos 2t}{x}$$

where x cm is the height of the cradle above its base t seconds after the cradle begins to oscillate.

Given that the cradle is 20 cm above its base at time $t = \frac{\pi}{4}$ seconds, find:

- (i) the height of the cradle above its base 13 seconds after it starts oscillating, giving your answer to the nearest millimetre;
- (ii) the time at which the cradle will first be 11 cm above its base, giving your answer to the nearest tenth of a second.

(2)

(2)

(Total 10 marks)

Q9.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - 7x - 6$.

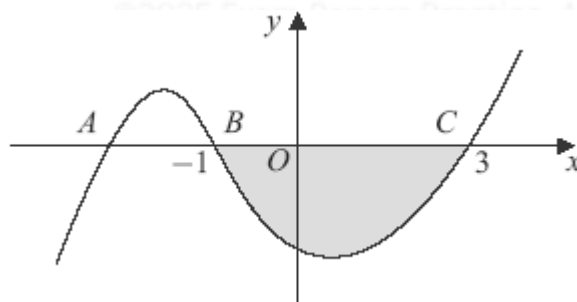
- (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

- (ii) Express $p(x) = x^3 - 7x - 6$ as the product of three linear factors.

(3)

- (b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



The curve cuts the x -axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

- (i) State the coordinates of the point A .

(1)

- (ii) Find $\int_{-1}^3 (x^3 - 7x - 6) dx$.

(5)

- (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x -axis between B and C .

(1)

- (iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B .

(3)

- (v) Hence find an equation of the normal to the curve at the point B .

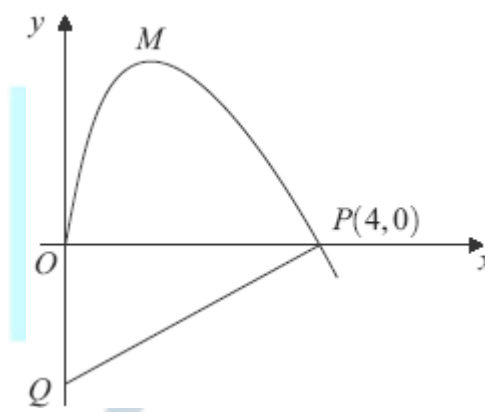
(3)

(Total 18 marks)

Q10.

A curve, drawn from the origin O , crosses the x -axis at the point $P(4, 0)$.

The normal to the curve at P meets the y -axis at the point Q , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = 4x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

- (a) (i) Find $\frac{dy}{dx}$.

(3)

- (ii) Show that the gradient of the curve at $P(4, 0)$ is -2 .

(2)

- (iii) Find an equation of the normal to the curve at $P(4, 0)$.

(3)

- (iv) Find the y -coordinate of Q and hence find the area of triangle OPQ .

(3)

- (v) The curve has a maximum point M . Find the x -coordinate of M .

(3)

(b) (i) Find $\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx$.

(3)

(ii) Find the total area of the region bounded by the curve and the lines PQ and QO .

(3)

(Total 20 marks)

Q11.

(a) (i) Given that $y = 2x^2 - 8x + 3$, find $\frac{dy}{dx}$.

(1)

(ii) Hence, or otherwise, find

$$\int_4^6 \frac{x-2}{2x^2-8x+3} dx$$

giving your answer in the form $k \ln 3$, where k is a rational number.

(4)

(b) Use the substitution $u = 3x - 1$ to find , giving your answer in terms of x .

(4)

(Total 9 marks)

Q12.

Solve the differential equation

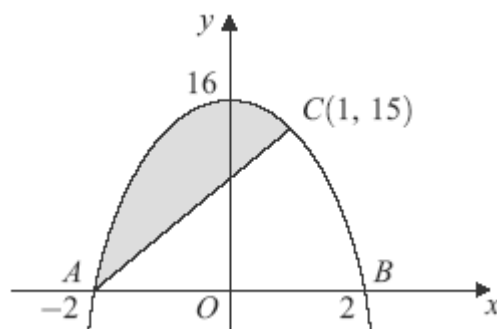
$$\frac{dy}{dx} = \frac{3 \cos 3x}{y}$$

given that $y = 2$ when $x = \frac{\pi}{2}$. Give your answer in the form $y^2 = f(x)$.

(Total 5 marks)

Q13.

The curve with equation $y = 16 - x^4$ is sketched below.



The points $A(-2, 0)$, $B(2, 0)$ and $C(1, 15)$ lie on the curve.

- (a) Find an equation of the straight line AC .

(3)

- (b) (i) Find $\int_{-2}^1 (16 - x^4) dx$.

(5)

- (ii) Hence calculate the area of the shaded region bounded by the curve and the line AC .

(3)

(Total 11 marks)

Q14.

- (a) The expression $\left(1 + \frac{4}{x^2}\right)^3$ can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^6}$$

By using the binomial expansion, or otherwise, find the values of the integers p and q .

(3)

- (b) (i) Hence find $\int \left(1 + \frac{4}{x^2}\right)^3 dx$.

(4)

- (ii) Hence find the value of $\int_1^2 \left(1 + \frac{4}{x^2}\right)^3 dx$.

(2)

(Total 9 marks)

Q15.

- (a) By writing $\sin 3x$ as $\sin(x + 2x)$, show that $\sin 3x = 3 \sin x - 4 \sin^3 x$ for all values of x .

(5)

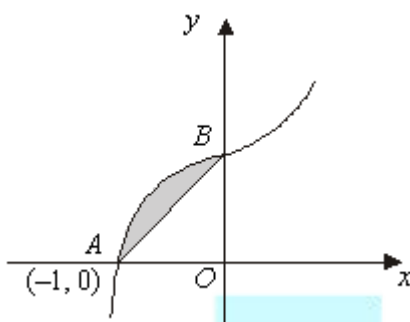
- (b) Hence, or otherwise, find $\int \sin^3 x dx$.

(3)

(Total 8 marks)

Q16.

The curve with equation $y = 3x^5 + 2x + 5$ is sketched below.



The curve cuts the x -axis at the point $A(-1, 0)$ and cuts the y -axis at the point B .

- (a) (i) State the coordinates of the point B and hence find the area of the triangle AOB , where O is the origin.

(3)

- (ii) Find $\int (3x^5 + 2x + 5) dx$

(3)

- (iii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

- (b) (i) Find the gradient of the curve with equation $y = 3x^5 + 2x + 5$ at the point $A(-1, 0)$.

(3)

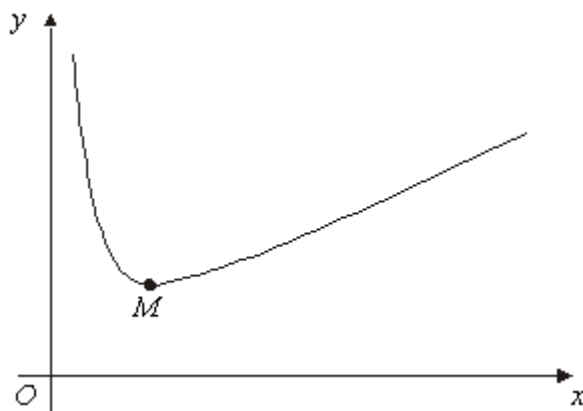
- (ii) Hence find an equation of the tangent to the curve at the point A .

(1)

(Total 14 marks)

Q17.

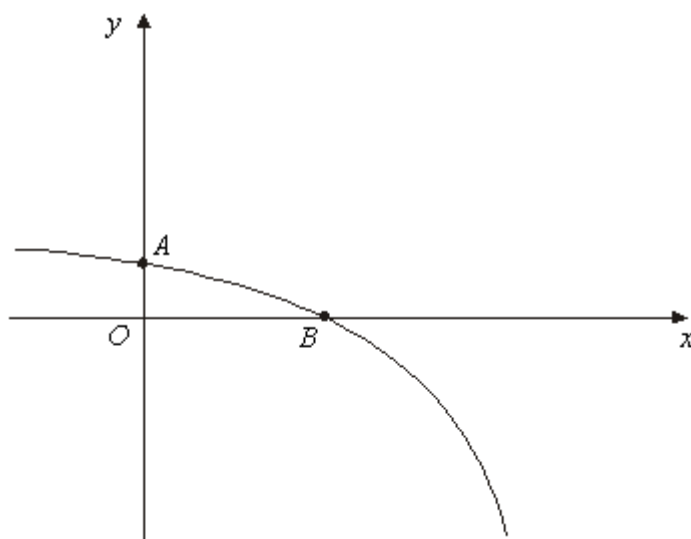
A curve C is defined for $x > 0$ by the equation $y = x + 1 + \frac{4}{x^2}$ and is sketched below.



- (a) (i) Given that $y = x + 1 + \frac{4}{x^2}$, find $\frac{dy}{dx}$. (3)
- (ii) The curve C has a minimum point M . Find the coordinates of M . (4)
- (iii) Find an equation of the normal to C at the point $(1, 6)$. (4)
- (b) (i) Find $\int \left(x + 1 + \frac{4}{x^2} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the lines $x = 1$ and $x = 4$ and the x -axis. (2)
- (Total 16 marks)

Q18.

The sketch shows the graph of $y = 4 - e^{2x}$. The curve crosses the y -axis at the point A and the x -axis at the point B .



- (a) (i) Find $\int (4 - e^{2x}) dx$. (2)
- (ii) Hence show that $\int_0^{\ln 2} (4 - e^{2x}) dx = 4 \ln 2 - \frac{3}{2}$. (2)
- (b) (i) Write down the y -coordinate of A . (1)
- (ii) Show that $x = \ln 2$ at B . (2)
- (c) Find the equation of the normal to the curve $y = 4 - e^{2x}$ at the point B . (4)
- (d) Find the area of the region enclosed by the curve $y = 4 - e^{2x}$, the normal to the curve at B and the y -axis. (3)
- (Total 14 marks)**

Q19.

- (a) Express $\cos 2x$ in terms of $\sin x$. (1)
- (b) (i) Hence show that $3 \sin x - \cos 2x = 2 \sin^2 x + 3 \sin x - 1$ for all values of x . (2)
- (ii) Solve the equation $3 \sin x - \cos 2x = 1$ for $0^\circ < x < 360^\circ$. (4)

- (c) Use your answer from part (a) to find $\int \sin^2 x \, dx$.

(2)

(Total 9 marks)

Q20.

- (a) (i) Solve the differential equation $\frac{dy}{dt} = y \sin t$ to obtain y in terms of t .

(4)

- (ii) Given that $y = 50$ when $t = \pi$, show that $y = 50e^{-(1 + \cos t)}$.

(3)

- (b) A wave machine at a leisure pool produces waves. The height of the water, y cm, above a fixed point at time t seconds is given by the differential equation

$$\frac{dy}{dt} = y \sin t$$

- (i) Given that this height is 50 cm after π seconds, find, to the nearest centimetre, the height of the water after 6 seconds.

(2)

- (ii) Find $\frac{d^2y}{dt^2}$ and hence verify that the water reaches a maximum height after π seconds.

(4)

(Total 13 marks)

Q21.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 + 4x - 5$. All rights reserved.

- (i) Use the Factor Theorem to show that $x - 1$ is a factor of $f(x)$

(2)

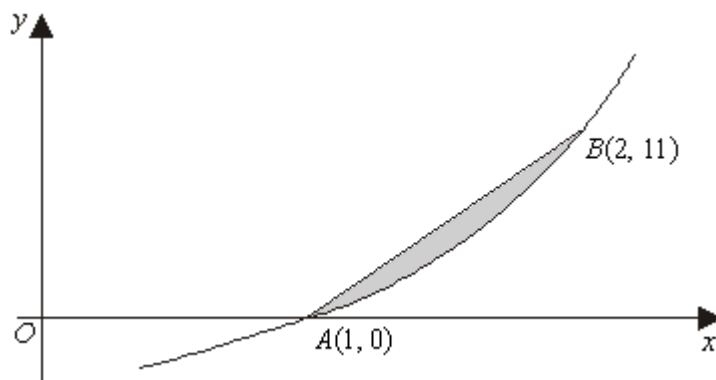
- (ii) Express $f(x)$ in the form $(x - 1)(x^2 + px + q)$, where p and q are integers.

(2)

- (iii) Hence show that the equation $f(x) = 0$ has exactly one real root and state its value.

(3)

- (b) The curve with equation $y = x^3 + 4x - 5$ is sketched below.



The curve cuts the x -axis at the point $A(1, 0)$ and the point $B(2, 11)$ lies on the curve.

(i) Find $\int (x^3 + 4x - 5) \, dx$.

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

(Total 14 marks)

Q22.

(a) Simplify:

(i) $x^{\frac{3}{2}} \times x^{\frac{1}{2}}$;

(1)

(ii) $x^{\frac{3}{2}} \div x$;

(1)

(iii) $\left(x^{\frac{3}{2}}\right)^2$

(1)

(b) (i) Find $\int 3x^{\frac{1}{2}} \, dx$.

(3)

(ii) Hence find the value of $\int_1^9 3x^{\frac{1}{2}} \, dx$.

(2)

(Total 8 marks)

Q23.

- (a) Use integration by parts to find $\int x e^{5x} dx$.

(4)

- (b) (i) Use the substitution $u = \sqrt{x}$ to show that

$$\int \frac{1}{\sqrt{x}(1+\sqrt{x})} dx = \int \frac{2}{1+u} du$$

(2)

- (ii) Find the exact value of $\int_1^9 \frac{1}{\sqrt{x}(1+\sqrt{x})} dx$

(3)

(Total 9 marks)

Q24.

- (a) Write down $\int \sec^2 x dx$.

(1)

- (b) Given that $y = \frac{\cos x}{\sin x}$, use the quotient rule to show that $\frac{dy}{dx} = -\operatorname{cosec}^2 x$.

(4)

- (c) Prove the identity $(\tan x + \cot x)^2 = \sec^2 x + \operatorname{cosec}^2 x$.

(3)

- (d) Hence find $\int_{0.5}^1 (\tan x + \cot x)^2 dx$ giving your answer to two significant figures.

(4)

(Total 12 marks)

Q25.

- (a) Solve the differential equation

$$\frac{dy}{dx} = \frac{\sqrt{1+2y}}{x^2}$$

given that $y = 4$ when $x = 1$.

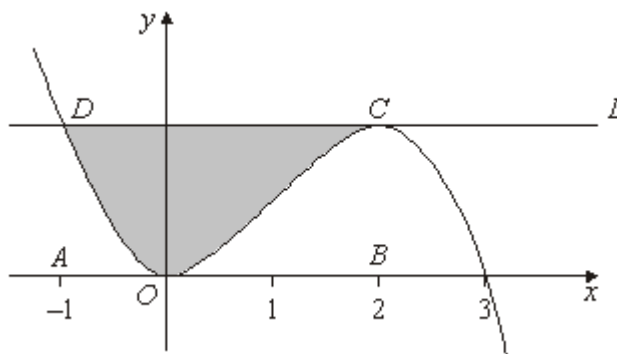
(6)

- (b) Show that the solution can be written as $y = \frac{1}{2} \left(15 - \frac{8}{x} + \frac{1}{x^2} \right)$

(2)
(Total 8 marks)

Q26.

The diagram shows the curve with equation $y = 3x^2 - x^3$ and the line L .

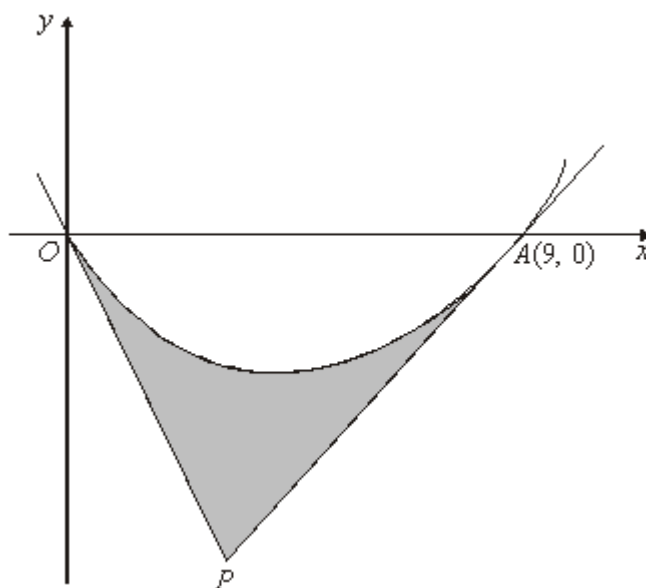


The points A and B have coordinates $(-1, 0)$ and $(2, 0)$ respectively. The curve touches the x -axis at the origin O and crosses the x -axis at the point $(3, 0)$. The line L cuts the curve at the point D where $x = -1$ and touches the curve at C where $x = 2$.

- (a) Find the area of the rectangle $ABCD$. (2)
- (b) (i) Find $\int (3x^2 - x^3) dx$. (3)
- (ii) Hence find the area of the shaded region bounded by the curve and the line L . (4)
- (c) For the curve above with equation $y = 3x^2 - x^3$: (2)
- (i) find $\frac{dy}{dx}$; (2)
- (ii) hence find an equation of the tangent at the point on the curve where $x = 1$; (3)
- (iii) show that y is decreasing when $x^2 - 2x > 0$. (2)
- (d) Solve the inequality $x^2 - 2x > 0$. (2)
- (Total 18 marks)

Q27.

A curve, drawn from the origin O , crosses the x -axis at the point $A(9, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = x^{\frac{3}{2}} - 3x$$

- (a) Find $\frac{dy}{dx}$. (2)

- (b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O . (2)

- (ii) Show that the equation of the tangent at $A(9, 0)$ is $2y = 3x - 27$. (3)

- (iii) Hence find the coordinates of the point P where the two tangents meet. (3)

- (c) Find $\int \left(x^{\frac{3}{2}} - 3x \right) dx$. (3)

- (d) Calculate the area of the shaded region bounded by the curve and the tangents OP and AP . (5)

(Total 18 marks)

Q28.

- (a) Express $\cos 2x$ in the form $a \cos^2 x + b$, where a and b are constants.

(2)

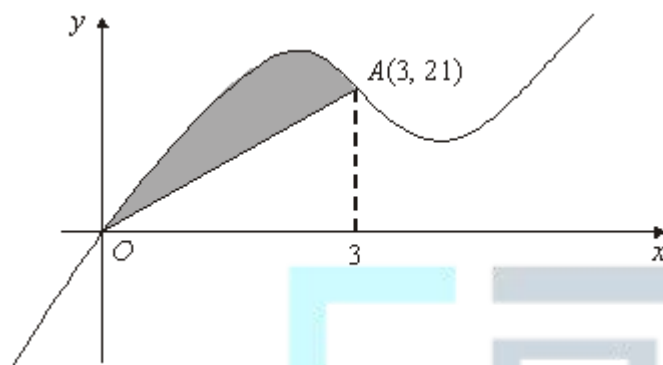
- (b) Hence show that $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{\pi}{a}$, where a is an integer.

(5)

(Total 7 marks)

Q29.

The curve with equation $y = x^3 - 10x^2 + 28x$ is sketched below.



The curve crosses the x -axis at the origin O and the point $A(3, 21)$ lies on the curve.

- (a) (i) Find $\frac{dy}{dx}$.

(3)

- (ii) Hence show that the curve has a stationary point when $x = 2$ and find the x -coordinate of the other stationary point.

(4)

- (b) (i) Find $\int (x^3 - 10x^2 + 28x) \, dx$.

(3)

- (ii) Hence show that $\int_0^3 (x^3 - 10x^2 + 28x) \, dx = 56 \frac{1}{4}$.

(2)

- (iii) Hence determine the area of the shaded region bounded by the curve and the line OA .

(3)

(Total 15 marks)

Q30.

At the point (x, y) , where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7$$

- (a) (i) Verify that $\frac{dy}{dx} = 0$ when $x = 4$. (1)

- (ii) Write $\frac{16}{x^2}$ in the form $16x^k$, where k is an integer. (1)

- (iii) Find $\frac{d^2y}{dx^2}$. (3)

- (iv) Hence determine whether the point where $x = 4$ is a maximum or a minimum, giving a reason for your answer. (2)

- (b) The point $P(1, 8)$ lies on the curve.
- (i) Show that the gradient of the curve at the point P is 12. (1)

- (ii) Find an equation of the normal to the curve at P . (3)

- (c) (i) Find $\int (3x^{\frac{1}{2}} + \frac{16}{x^2} - 7) dx$ (3)

- (ii) Hence find the equation of the curve which passes through the point $P(1, 8)$. (3)

(Total 17 marks)

Q31.

- (a) Given that $\frac{9x^2 - 6x + 5}{(3x-1)(x-1)}$ can be written in the form $3 + \frac{A}{3x-1} + \frac{B}{x-1}$, where A and B are integers, find the values of A and B . (4)

- (b) Hence, or otherwise, find $\int \frac{9x^2 - 6x + 5}{(3x-1)(x-1)} dx$.

(4)
(Total 8 marks)

Q32.

Solve the differential equation

$$\frac{dy}{dx} = 6xy^2$$

given that $y = 1$ when $x = 2$. Give your answer in the form $y = f(x)$.

(Total 6 marks)

Q33.

(a) The function f is defined for all values of x by $f(x) = x^3 - 3x^2 - 6x + 8$.

(i) Given that $f(1) = 0$ and $f(-2) = 0$, write down two linear factors of $f(x)$.

(2)

(ii) Hence express $x^3 - 3x^2 - 6x + 8$ as the product of three linear factors.

(2)

(b) The curve with equation $y = x^3 - 3x^2 - 6x + 8$ is sketched below.



(i) The curve intersects the y -axis at the point A . Find the y -coordinate of A .

(1)

(ii) The curve crosses the x -axis when $x = -2$, when $x = 1$ and also at the point B . Use the results from part (a) to find the x -coordinate of B .

(1)

(c) (i) Find $\int (x^3 - 3x^2 - 6x + 8) dx$

(4)

(ii) Hence find the area of the shaded region bounded by the curve and the x -axis.

(3)

(Total 13 marks)

Q34.

(a) Write $\sqrt{x}$ in the form x^k , where k is a fraction. (1)

(b) Hence express $\sqrt{x}(x-1)$ in the form $x^p - x^q$. (2)

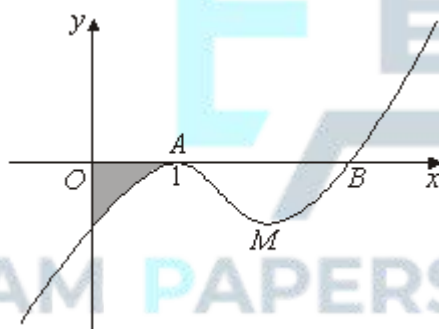
(c) Find $\int \sqrt{x}(x-1)dx$. (3)

(d) Hence show that $\int_1^2 \sqrt{x}(x-1)dx = \frac{4}{15}(\sqrt{2}+1)$. (3)

(Total 9 marks)

Q35.

The curve with equation $y = x^3 - 5x^2 + 7x - 3$ is sketched below.



The curve touches the x -axis at the point A (1, 0) and cuts the x -axis at the point B .

(a) (i) Use the factor theorem to show that $x - 3$ is a factor of $p(x) = x^3 - 5x^2 + 7x - 3$. (2)

(ii) Hence find the coordinates of B . (1)

(b) The point M , shown on the diagram, is a minimum point of the curve with equation $y = x^3 - 5x^2 + 7x - 3$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Hence determine the x -coordinate of M .

(3)

(c) Find the value of $\frac{d^2 y}{dx^2}$ when $x = 1$.

(2)

(d) (i) Find $\int (x^3 - 5x^2 + 7x - 3) dx$.

(4)

(ii) Hence determine the area of the shaded region bounded by the curve and the coordinate axes.

(4)

(Total 18 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.